Week 2 Progress

Bob Shen
Laidlaw Scholars Programme
Bishai Lab
July 2026

This is a team project

Hogan

- Weather and heat framing
- Goggins baseline challenge (Jan)
- Ages 65–69 and 70–74
- Climate X file and hot-month definitions

Roro

- Hospital Authority stroke aggregates
- Outcome timing and transfer path

Dr Bishai

- Original multi-method plan
- Teamwork: go far together
- Jasmine source confirmed; extend definitions to 2023

Also in the room this week

- Writing early (lit + methods) matters
- Equity lens (Marmot): who bears risk
- Peer heat work: summit, district voice, LegCo

This update credits the team's contributions. Judgments on weather and outcomes are still shared.

What the July meeting changed

Earlier plan	Meeting	Working now
AMI + stroke principal-dx	No admission reasons in general HA file	Stroke aggregates; AMI out of scope from that file
One primary heat model	Explore many methods	Labelled multi-spec panel
Heat metrics	Align with Ren/Wang-style spells	Spells, 2D3N, hot-month options with Hogan

Roro's outcome path (working understanding): first GOPC mention of stroke is a *marker*. The actual stroke and hospitalisation generally come earlier. We ignore later mentions and work back to the true event month.

The question we are preparing

How do **monthly** thermal exposures in Hong Kong, 2013–2023, associate with **stroke event aggregates**, and how stable is that picture across a shared panel of heat, cold, spell, and sensitivity specifications?

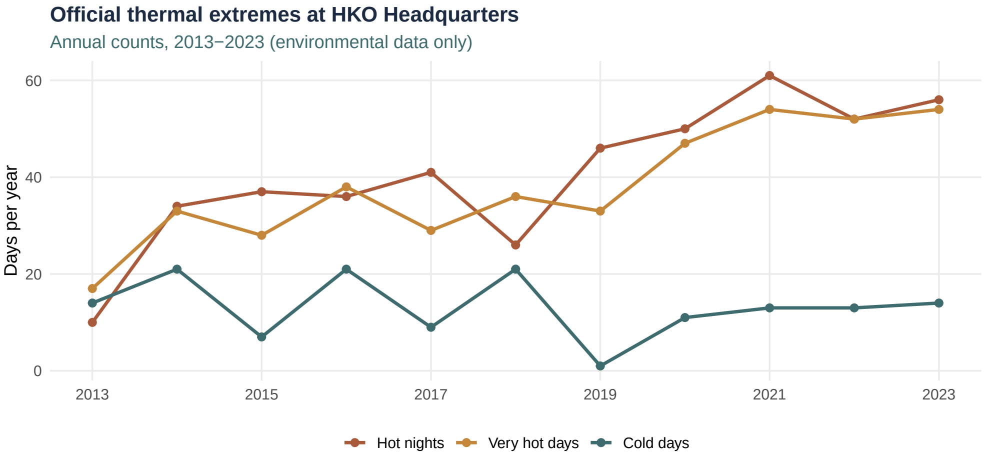
Complementary to daily heatwave–mortality work in the lab — not a duplicate. Not a daily DLNM; not AMI from a file without admission reasons.

From January: Hogan's intellectual lead

- Pulled Goggins MI (2013) and stroke (2012) as the Hong Kong baseline
- Warned that cold associations are already well documented — novelty must be careful
- Suggested comparing a later period (2013–2023) as an evolutionary question
- Highlighted ages **65–69** and **70–74** as actionable older bands
- Offered to prepare the monthly climate X file under Dr Bishai's plan
- Later: define heatwaves, count them by month, flag upper-tail "hot" months (Atmos. Res. heatwave framing; DOI 10.1016/j.atmosres.2024.107845)

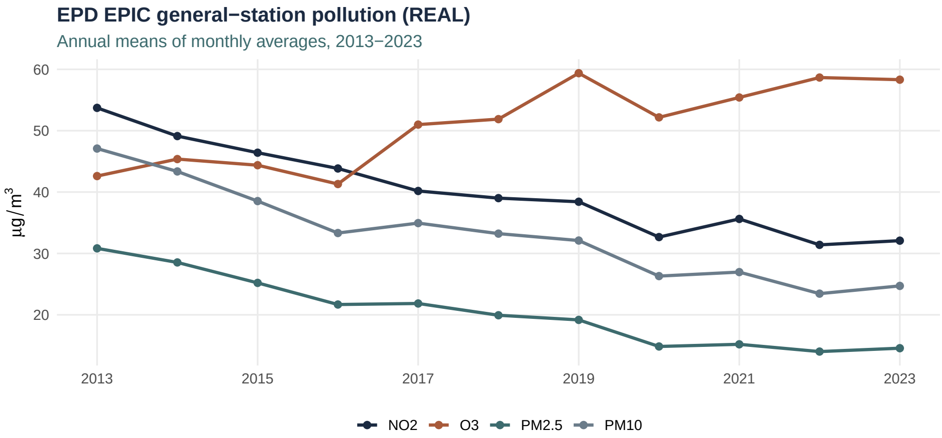
> Those contributions still structure the weather side. Definitions should be finished **with** Hogan, not around him.

Climate: shared ground truth (descriptive)



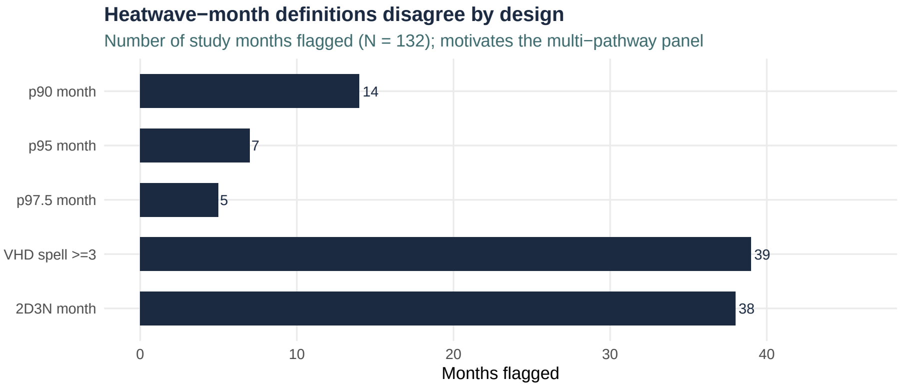
Hot nights and very hot days rose after 2018; cold days persist. Peak hot-night year in-window: **2021 (61)**. Environmental descriptives only — no stroke findings.

Pollution layer (EPD EPIC)



Official monthly general-station means: NO_2, O_3, $PM_{2.5}$, PM_{10}. NO_2/PM declined; O_3 rose — echoing the January discussion of a changing pollutant mix. Stage pollutants carefully; enter O_3 last.

Why heatwave definitions stay plural



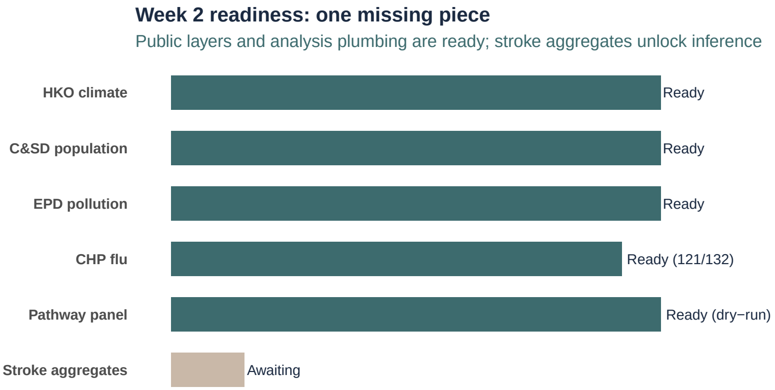
Guo/Gasparrini, Wang/Ren, and the lab's multi-definition mortality work all argue against one universal rule.
Hogan's hot-month idea (heatwave counts → upper-tail months) belongs here as a named option.

A labelled panel — not seventeen discoveries

Family	Examples (lit-anchored)
Continuous T / lag	Tmean; Tmax/Tmin; lag-1
Official extremes	Hot nights, very hot days, cold days (Hogan / HKO)
Spells & 2D3N	Ren/Wang-style burden metrics
Hot-month defs	Percentile months + Hogan count→tail idea
Cold & flu	Cold-primary; influenza co-exposure
Structure	Age bands Hogan flagged; sex if grain allows
Sensitivities	Pollution staging; humidity; COVID window

Headline proposal for Gate 3 *with the team*: continuous Tmax/Tmin + official extreme-day counts. Full pathway×literature map is in the Laidlaw write-up.

What each person still unlocks



Hogan: lock weather / hot-month / cold-month definitions, including Roro's four as named siblings.
Roro: first GOPC mention → true event month; transfer; revised PDF vs medRxiv v1.
Dr Bishai: thank you for identifying Jasmine as Jingwen Liu et al. (2020).
Public pollution and population layers can support the merge once outcomes arrive.

Honesty line

Ready to discuss

- Environmental descriptives
- Outcome construction logic with Roro
- Weather definitions with Hogan
- Lit + methods writing for Laidlaw

Not ready to claim

- Stroke coefficients
- “Heat replaced cold”
- AMI from this HA file
- Synthetic dry-runs as findings

Literature anchors (selected)

- **Goggins 2012/2013** — cold-dominant HK stroke/AMI baselines (Hogan's challenge)
- **Jingwen Liu et al. 2020** — confirmed Jasmine paper; cold $\gg$ heat mortality AF
- **Zhenyuan Liu et al. 2026** — team multi-definition excess-heat mortality baseline
- **Guo 2024** — hot-night intensity vs official binary flags
- **Wang/Ren 2019** — VHD/HN spells and 2D3N
- **Guo/Gasparrini 2017** — multi-definition heatwaves
- **Yang/Chong 2025** — cold + influenza and stroke
- **Guo 2025** — pollution can amplify temperature–hospitalisation risks

Jasmine is confirmed — cold $\gg$ heat burden

Jingwen Liu et al. (2020)
Cause-specific mortality attributable to cold and hot ambient temperatures in Hong Kong: a time-series study, 2006–2016.
Sustainable Cities and Society 57:102131.
DOI: `10.1016/j.scs.2020.102131`.

Attributable fraction	**Cold / moderate**	**Heat / extreme**
Thermal side	**Cold 4.72%**	Heat 0.16%
Intensity class	**Moderate 4.25%**	Extreme 0.63%

Daily mortality; DLNM + quasi-Poisson; reversed J-shaped curve. Extreme cold had the highest risk: injury RR 2.18 (1.03–4.62), lag 0–21.

One team evidence chain

Jasmine
Liu et al. 2020
daily mortality RR / AF
2006–2016

→ local RRs →

Roro
Liu et al. 2026
excess heat deaths
2014–2023

→ team baseline →

Our extension
monthly stroke aggregates
paired hot / cold
2013–2023

Roro paper: Zhenyuan Liu, Chao Ren, Jingwen Liu, Kawasaki Yurika and David Makram Bishai.
DOI: `10.64898/2026.03.05.26347683` (medRxiv v1).

Roro estimates 1,455 (1,098–1,812) to 3,238 (3,234–3,242) excess deaths across four heatwave definitions. Complementarity, not competition.

Roro's four-definition mortality baseline

Definition	Event / lag intent	RR source
HWD_Tavg	Mean T at 99th percentile (30.60 °C); 20-day window; lag phases 0, 1–5, 6–21	Jasmine
HWD_Tmax	Consecutive very hot days; $T_{\text{max}} \geq 33^{\circ}$C	Wang / Ren
HWD_Tmin	Consecutive hot nights; $T_{\text{min}} \geq 28^{\circ}$C	Wang / Ren
HWD_Tcombined	2 hot days + 3 hot nights (2D3N); event + five later days	Wang / Ren

The Mac `revised manuscript_clean.pdf` is not yet ingested. This is the medRxiv v1 baseline; exact run operators still need a source manifest.

Definition-harmonised extension through 2023

1. Ingest Jasmine's full PDF / supplement; lock reference, knots, lags and test universe.
2. Reconstruct daily hot and cold definitions without opening stroke associations.
3. Audit coding in the shared 2013–2016 window.
4. Apply unchanged definitions through December 2023.
5. Collapse daily burden to month; join to Roro's **true stroke-event month**.
6. Report exposed months, estimates and intervals; keep null counts secondary.

Daily mortality AF $\neq$ modelled excess deaths $\neq$ monthly stroke association. This is definition transport, not an exact replication.

Next week: symmetric hot and cold months

Hot families (HM)

- HKO hot nights / very hot days
- Monthly percentile tails
- VHD, HN and 2D3N spells
- Heatwave starts and event days
- Degree-day intensity
- Anomaly, compound, onset, warnings

Cold families (CM)

- HKO cold-day counts
- 12 °C and 10 °C sensitivities
- Monthly percentile tails
- Cold spells and event counts
- Cold degree-day intensity
- Flu, anomaly, warning, onset

Catalogue: 50 HM definitions, 48 CM definitions, and paired pathways H01–H12 / C01–C12.

Starter HM / CM set — run exposure audits first

Hot	**Rule**	**Cold**	**Rule**
HM23	Hogan p90 heatwave-count month	CM03	$\geq$ 5 days with $T_{\text{min}} \leq 12^{\circ}$C
HM08	Monthly mean temperature $\geq$p90	CM08	Monthly mean temperature $\leq$p05
HM15	Touches a 5-VHD spell	CM15	Touches a 3-day cold spell
HM17	Touches a 5-HN spell	CM05	Any $T_{\text{min}} \leq 10^{\circ}$C day
HM19	Touches a 2D3N event	CM30	Cold + influenza sensitivity

First-wave intensity / compound additions: HM27 night heat-degree burden and HM32 hot + O_3. Freeze thresholds, selected months and missingness before outcomes.

Hogan's atmospheric recipe is one HM family

Li et al.
May–September
$T_{\max} >$ calendar-day p90
$\geq$ 3 consecutive days

$\rightarrow$

Hogan adaptation
count event starts
within each month

$\rightarrow$

Flag upper-tail
monthly counts
HM23 / HM24

Li C et al., *Atmospheric Research* 315 (2025) 107845.
DOI: `10.1016/j.atmosres.2024.107845`.

Named and credited — beside HKO counts, Jasmine transport, Roro / Wang–Ren definitions, percentiles and cold counterparts.

Gene / ~10 °C: hypothesis only

- Discussion idea: South China populations may appear relatively heat-adapted, while temperatures around ~10 °C may remain consequential.
- This ecological monthly study cannot identify genes, ancestry effects or causal adaptation.
- Housing, cooling, behavior, occupation, age, health care and acclimatization remain alternative explanations.
- The 10 °C rule is a candidate CM sensitivity, not a biological cutoff.
- No stroke findings yet; synthetic outputs are not findings.

Small asks: source-table lock with Dr Bishai, weather-definition lock with Hogan, and true-event-month lock with Roro :)

Laidlaw Stage 3 is on the horizon

- Research essay (2000–3000 words) — lit + methods drafts are started
- A0 **portrait** poster (841×1189 mm) — research focus; results pending outcomes
- HKU report form + supervisor endorsement
- Showcase: HKU Laidlaw Society (early Nov 2026) and Annual Conference

Writing early — as other presenters modelled — is how we stay honest about what is ready and what still needs the team.

Thank you

Next small asks — whenever you have a moment :)

- **Hogan:** shall we lock the heatwave / hot-month definitions together (including your count→upper-tail idea, the cold starters, and Roro's four)?
- **Roro:** does `revised manuscript_clean.pdf` differ from medRxiv v1, and can we lock first-GOPC→true-stroke timing plus aggregate-transfer timing?

If you want to go fast, go alone; if you want to go far, go together.

Thank you, Dr Bishai, for the Jasmine ID; grateful to Hogan and Roro too.